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Three-month lower-dose flecainide after catheter ablation of atrial fibrillation

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Aims Atrial tachyarrhythmias (AT) commonly recur within the first 3 months after radiofrequency catheter ablation (RFCA) of atrial fibrillation (AF), and the influence of antiarrhythmic drugs (AADs) on the recurrences has not been fully elucidated. We sought to evaluate the efficacy of a 3-month lower-dose flecainide regime on early and late recurrences of ATs.

Methods and results We randomly assigned 126 patients, who underwent RFCA for AF, to the flecainide group (150 or 100 mg/day according to their body weight) or to the control group receiving no AADs. The primary endpoint was any AT lasting for $\geq$30 s during the first 3 months and the secondary endpoint was a composite of ATs lasting for $\geq$24 h or requiring cardioversion or hospitalization during the same period. All AADs were stopped after the first 3 months and the late arrhythmia recurrences were also evaluated. The primary endpoint rates were 37 and 41% in the flecainide (143 $\pm$ 19 mg/day) and control groups, respectively (log-rank $P = 0.76$), and those of the secondary endpoint were 10 and 14%, respectively (log-rank $P = 0.45$). The estimated rates of maintaining sinus rhythm at 12 months after the first 3 months were 78 and 72%, in the flecainide and control groups, respectively (log-rank $P = 0.68$), and the rates were 51 and 90% in those with and without the primary endpoint, respectively (log-rank $P < 0.001$).

Conclusion The 3-month lower-dose flecainide therapy after AF ablation did not reduce the early and late arrhythmia recurrences. The clinically significant ATs were also not prevented.

Keywords Atrial fibrillation • Radiofrequency catheter ablation • Flecainide • Antiarrhythmic drug

Introduction

Radiofrequency catheter ablation (RFCA) achieving pulmonary vein isolation and the elimination of other arrhythmic substrates has been proposed as a highly effective therapeutic option for patients with atrial fibrillation (AF).[1–5] Early recurrences of AF or atrial tachycardia (AT) following the ablation are common, but the patients with the early recurrences do not necessarily experience arrhythmias in the late period.[2,6–12] A recent expert consensus recommendation[13] endorsed a 'blanking period' of 3 months, during which the emergence of tachyarrhythmias should not be classified as a treatment failure.

On the other hand, the rate of late recurrences is significantly higher in those with early AF/AT recurrences than in those without.[2,8–10,12] A strategy for the suppression of early recurrences after AF ablation to improve the long-term outcomes, however, has not been fully established, and there are only a few randomized studies addressing the management of early recurrences by antiarrhythmic drugs (AADs). Roux *et al.*[7] showed that empirical therapy with various AADs for the first 6 weeks after AF ablation reduced the incidence of clinically significant atrial arrhythmias but the AAD therapy did not prevent late arrhythmia relapses.[14] In their study, the efficacy of individual AADs was not evaluated, the endpoint was not an emergence of any AF/AT episodes but a composite including atrial arrhythmias lasting for >24 h, the duration of the AAD therapy was much shorter than 3 months, and 25% of the patients had a history of a previous AF ablation. The present study was conducted to clarify the efficacy of flecainide, one of the most commonly used AAD, for the first 3 months after the initial AF ablation in regard to the incidence of any early and late AF/AT recurrences. A systematic review[15] of the prior studies showed significantly increased proarrhythmic effects and drug withdrawals in the AF patients treated with flecainide, so the present study adopted a lower-dose flecainide therapy.

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What's new?

- In the present prospective randomized trial including 126 patients who underwent an initial radiofrequency catheter ablation for atrial fibrillation (AF), the recurrence rates of AF or atrial tachycardia (AT) during the first 3 months after the ablation did not differ between the patients assigned to a 3-month lower-dose flecainide therapy (143 ± 19 mg/day) and those assigned to a no antiarrhythmic drug therapy (37 vs. 41%, respectively, log-rank $P = 0.76$).
- Early use of the lower-dose flecainide did not prevent the late AF/AT recurrences (26 vs. 29%, log-rank $P = 0.68$).
- The rates of AF/AT lasting ≥24 h or requiring cardioversion or hospitalization during the first 3 months also did not differ between the flecainide group and the no antiarrhythmic drug group (10 vs. 14%, respectively, log-rank $P = 0.45$). These results indicate that the lower-dose flecainide during the early period after AF ablation has only a limited role.

Methods

Study protocol

The present study was designed as a prospective randomized non-blinded trial. The study protocol was approved by the institutional review board at Nippon Medical School. All patients referred to the Nippon Medical School Teaching Hospital for ablation of AF were screened. Patients were excluded if they were <18 years; had a history of RFCA or surgery for AF; were on amiodarone therapy within 3 months or on bepridil therapy within 1 month; suffered from congestive heart failure or hypertrophic cardiomyopathy; showed any symptoms, electrocardiographic abnormalities, or images suggesting ischaemic heart disease; showed a left ventricular ejection fraction of <0.50; had a diagnosis of sick sinus syndrome; had any syncopal episodes; showed a resting heart rate of <50 beats per minute or second- or third-degree atrioventricular block; underwent haemodialysis; were unable to be followed-up at the study site; were participating in another clinical trial evaluating AAD efficacy; or were unwilling to provide informed consent. They also were excluded if all pulmonary veins were not isolated.

After the RFCA, eligible patients were randomized to either the flecainide or control group. The follow-up period after the RFCA was divided into the early and late periods defined as the first 3 months after the session and the term thereafter, respectively (*Figure 1*). In those assigned to the flecainide group, the drug was prescribed during the early period, and, in the late period, all patients were followed without any AADs until the development of at least one tachyarrhythmia episode. The primary endpoint of the present study was any episode of AF or AT lasting longer than 30 s during the early period, and the secondary endpoint was defined as a composite of the following: (i) any episode of AF or AT lasting for >24 h; and (ii) severely symptomatic or haemodynamically deleterious ATs requiring cardioversion or emergency admission during the early period. The incidence of late recurrences, which was also defined as any AF/AT episodes lasting longer than 30 s, was also compared between the flecainide and control groups. The data concerning an emergence of bradyarrhythmias or ventricular tachyarrhythmias were also analysed. Further, as a surrogate measure of the arrhythmia burden among the patients experiencing AF/AT recurrences during the early period, the ratio of the number of days in which any AF/ATs longer than 30 s were recorded to the total number of days under flecainide therapy in the flecainide group and no AAD therapy in the control group was calculated, and compared between the groups.

Flecainide therapy after the ablation

In the patients randomly assigned to the lower-dose flecainide group, the drug was started on the evening after the ablation procedure, and administered b.i.d. with a daily dose of 150 mg in those weighing ≥50 kg and 100 mg in those weighing <50 kg. The treating electrophysiologists (M. H. and Y. M.) continued flecainide with the same dose during the early period unless there were observable side effects, such as symptomatic hypotension (systolic blood pressure of <90 mm Hg), bradycardia during sinus rhythm (heart rate of <50 beats per minute) or after restoration of sinus rhythm from AF (sinus pause of >4 s), dyspnoea, presyncope, syncope, or episodes of ventricular tachyarrhythmia developed. Flecainide was changed to another AAD if there were at least three episodes of AF/AT or the development of the secondary endpoint, which was a clinically significant AT. The choice of the alternative AAD depended on the treating electrophysiologists. In the patients assigned to the control group, the administration of Class I or III AADs was avoided unless there was a patient request to take AADs, at least three episodes of ATs, or an occurrence of the secondary endpoint.

Catheter ablation procedure

Before the RFCA, all AADs were discontinued for at least five half-lives. Two 7-Fr duo-decapolar circular mapping catheters (Lasso, Biosense Webster), and a 4 mm tip irrigated ablation catheter (Safire BLU Duo) were inserted into the left atrium. Procedures were guided using an electroanatomical mapping system (Ensite NavX). Bipolar endocardial electrograms were monitored continuously, and recorded on an EP-WorkMate recording system (EP MedSystems) with a filter setting of 30 −500 Hz. Ablation was performed by the delivery of radiofrequency current between the distal electrode of the mapping catheter and a cutaneous adhesive electrode patch, targeting a maximum temperature of 41°C and a maximum power of 30 W. A manually controlled steerable sheath (Agilis) was used in all procedures. Continuous radiofrequency lesions surrounding each ipsilateral pulmonary vein antrum were deployed. Two circular mapping catheters were positioned inside the ipsilateral superior and inferior pulmonary veins to verify pulmonary vein isolation, which was considered successful when no pulmonary vein potentials were recorded along the antrum or inside the vein and an exit block of spontaneous pulmonary vein firing or pacing at several sites within the pulmonary vein antrum was confirmed. Although the creation of additional lesions depended on the operator's decision, the standard AF ablation strategy in our laboratory was pulmonary vein isolation, cavotricuspid isthmus conduction block, and left posterior atrial wall isolation by creating roof and posterior inferior linear lesions which connected the superior and inferior aspects of the septal and lateral pulmonary veins.[5] The isolation of the posterior wall was confirmed by the disappearance of any atrial potentials inside the posterior left atrium during sinus rhythm and exit block of the pacing delivered at several sites within there. A mitral isthmus linear ablation[4] was added, especially in the patients with persistent AF or those with a left atrial dimension of >50 mm.

Post-ablation follow-up

After the RFCA, the patient's heart rhythm was continuously recorded with telemetry monitoring (DS-7680W, Fukuda Denshi,) until discharge from the hospital, and, then, was recorded with a cardiac event recorder (OMRON HCG-901, Omron Healthcare). The cardiac event recorder was utilized in all patients for 4 months after the ablation: for the entire early period and the first month of the late period. The patients were

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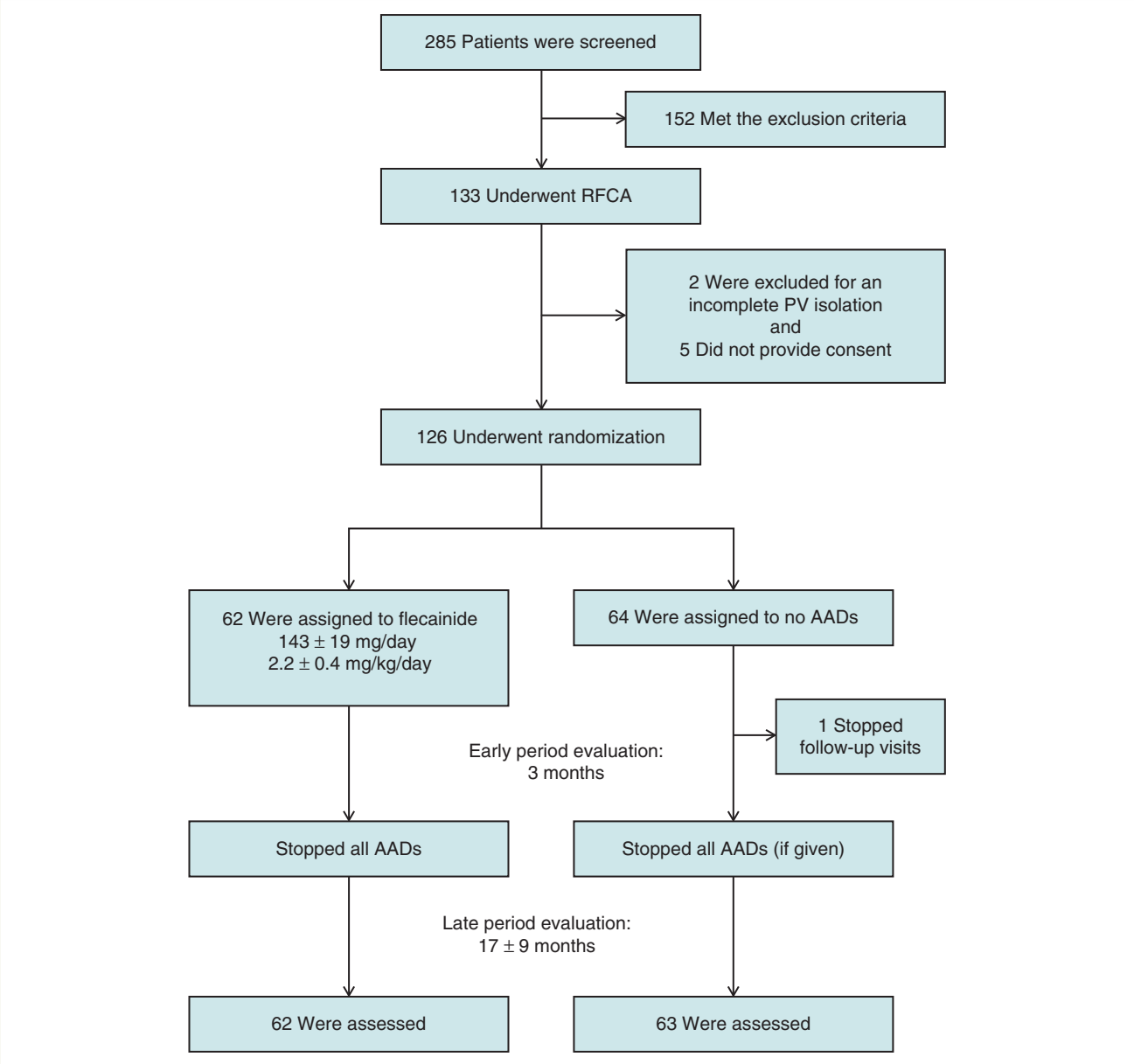

Figure 1 Screening, randomization, and follow-up. AAD, antiarrhythmic drug; PV, pulmonary vein; RFCA, radiofrequency catheter ablation.

instructed to use it at least twice a day for 30 s each and an additional recording was encouraged in case of feeling any symptoms compatible with arrhythmias. The data of the cardiac event recorder were retrieved and examined at the outpatient clinic every visit. The patients were seen for a follow-up at 2 weeks, 1, 2, 3, 4, 5, 6, 8, 10, and 12 months after the procedure and every 3 months thereafter. A 12-lead electrocardiogram was performed every visit, and a 24 h Holter monitoring was evaluated at 6 and 12 months after the ablation. The patients were advised to come to the hospital and undergo an additional electrocardiogram in the event of palpitations, dyspnoea, near syncope, or syncope.

Statistical analysis

The incidence of AT recurrences for the first 3 months post ablation is reported to range from 40 to 65%.[10–12,16] Assuming an incidence of recurrence of 50% in the control group, we estimated that the enrollment of 124 patients undergoing AF ablation would be necessary in order for the study to achieve a power of 0.80 to detect a reduction of 50%[7] in the incidence of early AF/AT recurrences in the flecainide group with a two-sided alpha level of 0.05.

The data are expressed as the mean ± standard deviation for continuous variables, and as absolute frequencies and percentages for categorical variables. For continuous variables and categorical variables, differences between groups were compared using Student's *t*-test and Fisher's exact test, respectively. The time to the primary and secondary endpoints after AF ablation and to the late arrhythmia recurrence after the completion of the early period follow-up were expressed with the use of Kaplan–Meier curves and assessed with the log-rank test. The follow-up data were censored if the patient died or was lost to follow-up without an occurrence of an AF recurrence. Analyses were performed according to the intention-to-treat principle. All tests were two-sided and a *P* value of <0.05 was considered statistically significant. All statistical analyses were conducted using SPSS for Windows 11.0 J software (SPSS, Inc.).

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Table 1 Baseline characteristics of the study subjects

	Flecainide group (n = 62)	Control group (n = 64)	P value
Age, years	62 ± 11	64 ± 10	0.20
Male gender, n	48 (77%)	49 (77%)	1.00
Body weight, kg	67 ± 13	65 ± 12	0.37
Body mass index, kg/m^2	24.1 ± 2.9	23.8 ± 3.3	0.58
History of AF, years	5.2 ± 5.8	4.4 ± 3.5	0.33
AF type			
Paroxysmal AF, n	44 (71%)	46 (72%)	1.00
Persistent AF, n	18 (29%)	18 (28%)	1.00
Lasting duration, m	14 ± 16	19 ± 22	0.49
Concomitant common AFL, n	13 (21%)	15 (23%)	0.83
Number of failed AADs, n	1.0 ± 1.2	1.0 ± 1.0	0.75
Echocardiographic parameters			
Left atrial diameter, mm	38 ± 5	38 ± 6	0.91
LVEF, %	69 ± 8	68 ± 9	0.60
Comorbidities			
History of a stroke, n	5 (8%)	4 (6%)	0.74
Hypertension, n	40 (65%)	39 (61%)	0.72
Sleep apnoea, n	5 (8%)	3 (5%)	0.49
Diabetes mellitus, n	3 (5%)	6 (9%)	0.49
COPD, n	0	1 (2%)	1.00
Serum BNP, pg/mL	101 ± 96	82 ± 91	0.26

AAD, antiarrhythmic drug; AF, atrial fibrillation; AFL, atrial flutter; BNP, brain natriuretic peptide; COPD, chronic obstructive pulmonary disease; LVEF, left ventricular ejection fraction.

Results

Study subjects

Enrollment of the patients was started in November 2010, and 285 consecutive patients who were referred for AF ablation were screened until February 2013. Among them, 159 patients met predefined exclusion criteria: a history of RFCA for AF in 42, amiodarone therapy within 3 months in 4, bepridil therapy within 1 month in 5, congestive heart failure in 3, hypertrophic cardiomyopathy in 15, ischaemic heart disease in 5, left ventricular ejection fraction of <0.50 in 2, sick sinus syndrome in 13, undergoing haemodialysis in 3, patient refusal in 5, living too far from the study site to be followed in 2, enrollment in another clinical trial in 58, and failure to complete pulmonary vein isolation in 2. Overall, 126 patients [63 ± 11 years, 97 male patients (77%), 90 (71%) with paroxysmal AF] were randomized, and the flecainide and control groups consisted of 62 and 64 patients, respectively. *Table 1* summarizes the characteristics of the two groups, which were almost similarly distributed. The mean body weight of the patients assigned to the flecainide group was 67 ± 13 kg, and the average daily dose of the flecainide was 143 ± 19 mg, which was 2.2 ± 0.4 mg/kg.

Ablation lesions

After confirming the isolation of all pulmonary veins, linear lesions at the left atrial roof and posterior inferior wall were created in 120 patients (95%) and an isolation of the left atrial posterior wall was confirmed in 114 patients (90%) (*Table 2*). Mitral isthmus linear ablation was performed in 38 patients (30%) and the bidirectional conduction block was created in 34 patients (27%). There was no difference in the rate of a left posterior wall isolation between the flecainide and control groups (94 vs. 88%, $P = 0.36$), nor the rate of conduction block at the mitral isthmus (29 vs. 25%, $P = 0.69$) (*Table 2*). In all patients, a conduction block in the cavotricuspid isthmus was successfully created. At the end of the RFCA procedure, all patients were in sinus rhythm as a result of the ablation or electrical cardioversion.

Follow-up results during the early period

During the early period, all patients except for one in the control group who stopped coming to the hospital 2 months after the RFCA completed the scheduled follow-up. The primary endpoints were observed in 23 and 26 patients randomized to the flecainide and control groups, respectively (37 vs. 41%, log-rank $P = 0.76$) (*Figure 2A*): AF was seen in 19 (31%) and 20 (31%) patients, respectively (log-rank $P = 0.95$), and AT was recorded in 15 (24%) and 12 (19%) patients, respectively (log-rank $P = 0.51$). Considering the loading of the flecainide, another analysis for the primary endpoint was made, in which any AF/AT recurrences during the first 7 days after the ablation was ignored. The results also showed no difference between the two groups; the primary endpoint developed in 18 (29%) and 22 (34%) patients in the flecainide and control groups, respectively (log-rank $P = 0.63$). The secondary endpoints were seen in six and nine patients assigned to the flecainide and control

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Table 2 Ablation procedures

	Flecainide group (n = 62)	Control group (n = 64)	P value
Pulmonary vein isolation, n	62 (100%)	64 (100%)	1.00
Isolation of all pulmonary veins, n	62 (100%)	64 (100%)	1.00
Roof and posterior inferior linear lesions, n	61 (98%)	59 (92%)	0.21
Isolation of the left atrial posterior wall, n	58 (94%)	56 (88%)	0.36
Mitral isthmus linear ablation, n	20 (32%)	18 (28%)	0.70
Mitral isthmus conduction block, n	18 (29%)	16 (25%)	0.69
CFAE ablation, n	6 (10%)	8 (13%)	0.78
Superior vena cava isolation, n	3 (5%)	3 (5%)	1.00
Cavotricuspid isthmus conduction block, n	62 (100%)	64 (100%)	1.00

CFAE, complex fragmented atrial electrogram.

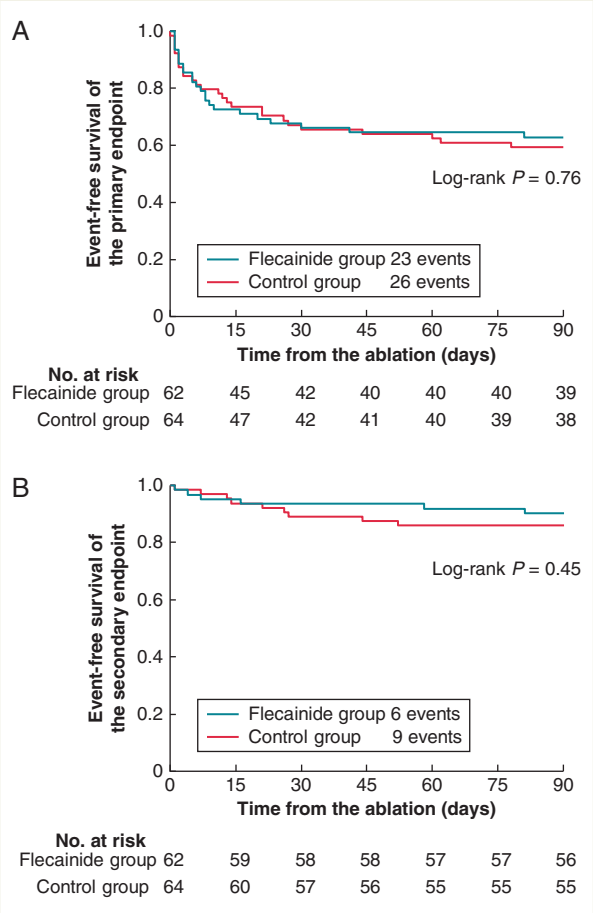

Figure 2 Kaplan–Meier curves during the 3-month post-ablation period. Shown is the time to the primary endpoint defined as any episode of AF or AT lasting longer than 30 s (*A*) and to the secondary endpoint, which was a composite of any AT lasting for >24 h or severely symptomatic or haemodynamically deleterious ATs requiring cardioversion or emergency admission (*B*), in the flecainide and control groups.

groups, respectively (10 vs. 14%, log-rank $P = 0.45$) (*Figure 2B*): AF/AT lasting for >24 h was seen in five and eight patients, respectively (log-rank $P = 0.42$), electrical cardioversion was performed in four and one patients, respectively (log-rank $P = 0.18$), and emergency admission occurred in one patient in the control group (log-rank $P = 0.33$). Among the patients with arrhythmia recurrences, the ratio of the number of days in which any AF/ATs longer than 30 s were recorded to the total number of days under flecainide therapy in the flecainide group and no AAD therapy in the control group was 0.23 ± 0.26 and 0.33 ± 0.34, respectively, and exhibited no statistical difference ($P = 0.19$).

Among those randomized to the flecainide group, the drug was stopped or changed to another AAD in 12 patients (19%) after the occurrence of AF/AT episodes in 5, AT requiring cardioversion in 2, bradyarrhythmias in 4 (2 of them also showed AF/AT recurrences), and drug-induced thrombocytopaenia in one who also experienced arrhythmia recurrences at that time. In the control group, an AAD therapy was started in 17 patients (27%) after AF/AT episodes in 15, and by the patient's request for the fear of an arrhythmia recurrence in 2 who actually did not experience any AF/AT episodes. Symptomatic bradycardia was seen in three and one patients in the flecainide and control groups, respectively (5 vs. 2%, $P = 0.36$). Ventricular tachyarrhythmias were not recorded in any of the study subjects.

Long-term outcomes after stopping antiarrhythmic drugs

In 125 patients who completed the early period evaluation, a late period follow-up without using any AADs was carried out. The mean follow-up period after starting the late period evaluation was 17 ± 9 months (range 4 to 31 months). Late recurrences of ATs were seen in 16 (26%) and 18 patients (29%) randomized to the flecainide and control groups, respectively. The estimated rates of maintaining sinus rhythm at 12 months after starting the late period were 78 and 72%, in the flecainide and control groups, respectively (log-rank $P = 0.68$) (*Figure 3A*). Also, the rates were 51 and 90% in

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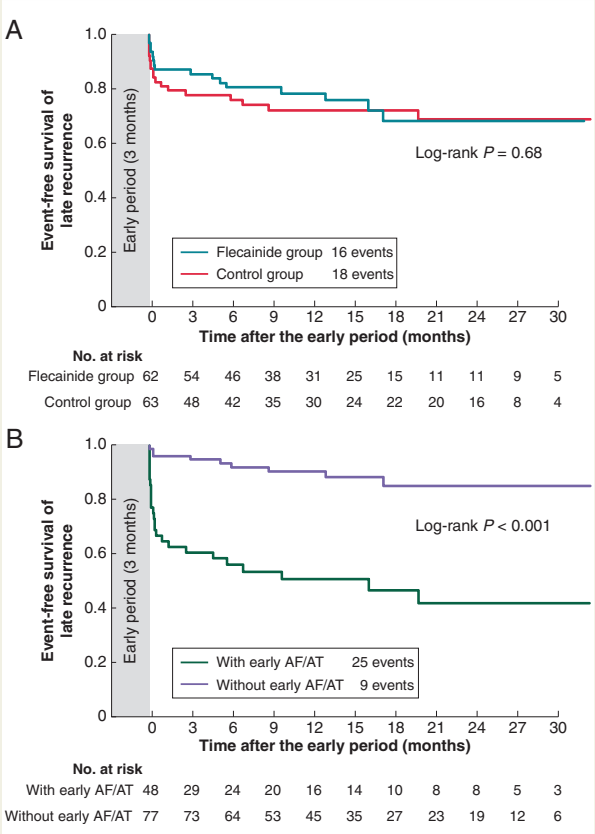

Figure 3 Kaplan–Meier curves during the late follow-up period. The late arrhythmia recurrence rates were compared between the flecainide and control groups (*A*), and between the patients with early arrhythmia recurrences and those without (*B*). AF, atrial fibrillation; AT, atrial tachycardia.

those with and without early AF/AT recurrences, respectively (log-rank $P < 0.001$) (*Figure 3B*).

Discussion

Early relapse of atrial tachyarrhythmias after the catheter ablation

In the present study using event recorders for an arrhythmia detection, AF/AT recurrences during the initial 3 months post RFCA, which was the primary endpoint, were recorded in 37 and 41% of the patients randomly assigned to the lower-dose flecainide therapy and no AAD therapy, respectively, yielding no statistical difference. The recurrence rate seen in our study subjects was almost similar to that reported in some prior studies.[10–12] These studies[10–12] also reported that up to 50% of the patients experiencing such early AF/AT recurrences did not show any further arrhythmias in the later follow-up period. The results of the present study were in agreement with these data; among a total of 48 patients with early arrhythmia recurrences, 23 (48%) were free from further AF/AT episodes during the late period without any AADs. Such divergence in the arrhythmia recurrence rates between the early and late periods after the RFCA is attributable to the difference in the arrhythmia mechanisms. The proposed mechanisms of early recurrences are mechanical and thermal injury provoking an inflammatory response[6,17,18] and autonomic nervous system modification,[19,20] but the recurrences in the later period can be mainly ascribed to the recovery of electrical conduction between the pulmonary veins and left atrium, or that of other block lines created in the previous procedure.[3,8,9,21] Studies demonstrating the favorable efficacy of a 3-day corticosteroid[6] and 3-month colchicine therapy[22] after AF ablation on the early AF recurrences support these theories.

Suppression of the early recurrences with antiarrhythmic drugs

Although the reason for no superiority of the flecainide group over the control group shown in the present study is not fully explainable, we consider the above-mentioned arrhythmia mechanisms early after the RFCA are the probable candidates. Flecainide is the most commonly used AAD for the suppression of paroxysmal AF.[23] Further, in the systematic review[15] and a recently reported prospective randomized study,[24] the administration of flecainide prevented arrhythmia recurrences more often after electrical cardioversion of persistent AF. Flecainide is considered appropriate in these settings for its sodium and potassium channel blocking effect as well as the induction of postrepolarization refractoriness;[25] however, it is unclear whether these antiarrhythmic properties can successfully cope with AF provoked by an inflammatory response or modified autonomic tone after ablation.

Randomized studies evaluating the efficacy of AADs on early recurrences after AF ablation have been scarce. One study[7] showed a lesser occurrence of composite endpoints, consisting of AF/AT lasting for >24 h, AF/AT requiring a hospital admission, cardioversion, or the initiation/change in the AAD therapy, and intolerance to AADs requiring drug cessation, in the patients treated with various empirically prescribed AADs than those treated without AADs (19 vs. 42%), but both the recurrence rate of any AT, and the efficacy of individual AADs were not provided. In that study, trans-telephonic monitoring was used only once a day and for 30 days, which might not be sufficient to detect early arrhythmia recurrences after RFCA. Further, 25% of the study subjects had a history of a previous AF ablation. In the repeat RFCA procedures, patients often require only a few touch-up lesions to isolate the pulmonary veins, which may provoke a minimal inflammatory response or autonomic nervous system modification.

In the present study, not only the rate of the primary endpoint, but also that of the secondary endpoint, which was a clinically significant arrhythmia similarly defined as the primary endpoint of the above-mentioned study,[7] did not differ between the flecainide and control groups. Even in the control group, the secondary endpoints were seen only in 14% of the patients. This result can be attributed to the relatively extensive ablation method; not only pulmonary vein isolation, but also left posterior wall isolation[5] and cavotricuspid isthmus conduction block was achieved in most of our study participants and mitral isthmus conduction block was also achieved in almost 30% of them. We consider that more extensive linear lesions might lead to greater inflammation or autonomic nerve modification, but

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diminish the provoked AF duration. In the institutes endorsing an extensive ablation strategy, there might be only a limited role of AADs to prevent clinically significant arrhythmias in the early period. In other words, AADs during the early period might be useful in patients undergoing AF ablation that consists of only pulmonary vein isolation and the elimination of non-pulmonary vein triggers.[7]

Regarding the extensive ablation strategy in our institute, its purpose was to suppress the arrhythmia recurrences as much as possible. Although it was difficult to prove its efficacy, the AF/AT recurrence rate without AADs at 1 year after the single ablation procedure shown in *Figure 3* was much lower than that seen in a meta-analysis of 63 AF ablation studies.[26] It, however, should be stressed that, when applying linear lesions, the most important thing is to create a complete conduction block, which was achieved in 95% of the present study subjects in whom left atrial roof and posterior inferior linear lesions were created (*Table 2*); otherwise they may become proarrhythmic.[27]

Lower-dose flecainide therapy in patients with atrial fibrillation

The daily dose of flecainide administered in the present study was 143 ± 19 mg. The approved daily dosage of flecainide for patients with AF in the USA and many European countries is 200–300 mg, and recent clinical studies[7,24,28] conducted in those countries utilized a mean daily dose of almost 200 mg. The primary reason for utilizing a lower daily dose in the present study was to avoid adverse side effects, such as proarrhythmias, significant bradycardia, and sudden cardiac death, which were shown in a prior systemic review[15] or a recent observational cohort study.[28] Other reasons were the approved daily dosage of flecainide in Japan, which was 100–200 mg, and the fact that Japanese physicians usually use lower-dose flecainide.[29] The daily dose of the flecainide prescribed in a highly specialized Japanese national medical centre treating cardiovascular and cerebrovascular diseases was reported to be 135.8 ± 47.4 mg.[30] Furthermore, the daily dose divided by the patient body weight may not quite differ between AF patients in Japan and those in the USA or European countries, because, as shown in *Table 1* and other studies,[24,30,31] the body weight of the former is around 65 kg, which is almost 20 kg lighter than the latter. It is, thus, suggested that the non-superiority of the flecainide in the primary and secondary endpoints in the present study was not fully ascribed to the lower dose.

Study limitations

This study has several limitations. First, we used lower-dose flecainide, and the efficacy of the flecainide with a daily dose of >200 mg is still uncertain. Further study evaluating the efficacy and safety of this drug in the patients undergoing AF ablation is warranted. Secondly, complete arrhythmia detection was not insured, as we did not implant a loop recorder in any of the study subjects. We, however, considered that a cardiac event recorder used at least twice a day and every time they had symptoms for 4 months after the RFCA was an appropriate measure to compare the arrhythmia recurrence between those with and without AADs. What we evaluated in the present study was not the absolute arrhythmia recurrence rate but the difference in the incidence between the two groups. We believe that the issue of arrhythmia detection should not have affected the main results of the present study. Thirdly, the extensive ablation strategy, which is not usual in the patients with paroxysmal AF in the other electrophysiology laboratories, might decrease the necessity for AAD therapy. The results of the present study should be validated in a future study in patients only undergoing pulmonary vein isolation.

Conclusion

In the present prospective randomized study including 126 patients who underwent an initial AF ablation, 3-month lower-dose flecainide therapy, compared with no AAD therapy, did not reduce the rate of early AF/AT recurrences nor that of the clinically significant ATs. The late arrhythmia recurrence rate also did not differ between the two groups. These results indicate that antiarrhythmic drugs might have only a limited role during the early period after AF ablation.

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References

1. Haissaguerre M, Shah DC, Jais P, Hocini M, Yamane T, Deisenhofer I *et al.* Electrophysiological breakthroughs from the left atrium to the pulmonary veins. *Circulation* 2000;**102**:2463–5.
2. Oral H, Knight BP, Tada H, Ozaydin M, Chugh A, Hassan S *et al.* Pulmonary vein isolation for paroxysmal and persistent atrial fibrillation. *Circulation* 2002;**105**:1077–81.
3. Jais P, Cauchemez B, Macle L, Daoud E, Khairy P, Subbiah R *et al.* Catheter ablation versus antiarrhythmic drugs for atrial fibrillation: the a4 study. *Circulation* 2008;**118**:2498–505.
4. Jais P, Hocini M, Hsu LF, Sanders P, Scavee C, Weerasooriya R *et al.* Technique and results of linear ablation at the mitral isthmus. *Circulation* 2004;**110**:2996–3002.
5. Sanders P, Hocini M, Jais P, Sacher F, Hsu LF, Takahashi Y *et al.* Complete isolation of the pulmonary veins and posterior left atrium in chronic atrial fibrillation. Long-term clinical outcome. *Eur Heart J* 2007;**28**:1862–71.
6. Koyama T, Tada H, Sekiguchi Y, Arimoto T, Yamasaki H, Kuroki K *et al.* Prevention of atrial fibrillation recurrence with corticosteroids after radiofrequency catheter ablation: a randomized controlled trial. *J Am Coll Cardiol* 2010;**56**:1463–72.
7. Roux JF, Zado E, Callans DJ, Garcia F, Lin D, Marchlinski FE *et al.* Antiarrhythmics after ablation of atrial fibrillation (5Astudy). *Circulation* 2009;**120**:1036–40.
8. Lee SH, Tai CT, Hsieh MH, Tsai CF, Lin YK, Tsao HM *et al.* Predictors of early and late recurrence of atrial fibrillation after catheter ablation of paroxysmal atrial fibrillation. *J Interv Card Electrophysiol* 2004;**10**:221–6.
9. Choi JI, Pak HN, Park JS, Kwak JJ, Nagamoto Y, Lim HE *et al.* Clinical significance of early recurrences of atrial tachycardia after atrial fibrillation ablation. *J Cardiovasc Electrophysiol* 2010;**21**:1331–7.
10. Themistoclakis S, Schweikert RA, Saliba WI, Bonso A, Rossillo A, Bader G *et al.* Clinical predictors and relationship between early and late atrial tachyarrhythmias after pulmonary vein antrum isolation. *Heart Rhythm* 2008;**5**:679–85.
11. Andrade JG, Macle L, Khairy P, Khaykin Y, Mantovan R, Martino GD *et al.* Incidence and significance of early recurrences associated with different ablation strategies for af: A STAR-AF substudy. *J Cardiovasc Electrophysiol* 2012;**23**:1295–1301.
12. Bertaglia E, Stabile G, Senatore G, Zoppo F, Turco P, Amellone C *et al.* Predictive value of early atrial tachyarrhythmias recurrence after circumferential anatomical pulmonary vein ablation. *Pacing Clin Electrophysiol* 2005;**28**:366–71.
13. Calkins H, Kuck KH, Cappato R, Brugada J, Camm AJ, Chen SA *et al.* 2012 HRS/EHRA/ECAS expert consensus statement on catheter and surgical ablation of atrial fibrillation: recommendations for patient selection, procedural techniques, patient management and follow-up, definitions, endpoints, and research trial design. *Europace* 2012;**14**:528–606.

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14. Leong-Sit P, Roux JF, Zado E, Callans DJ, Garcia F, Lin D *et al.* Antiarrhythmics after ablation of atrial fibrillation (5A study): six-month follow-up study. *Circ Arrhythm Electrophysiol* 2011;**4**:11–14.
15. Lafuente-Lafuente C, Mouly S, Longas-Tejero MA, Mahe I, Bergmann JF. Antiarrhythmic drugs for maintaining sinus rhythm after cardioversion of atrial fibrillation: a systematic review of randomized controlled trials. *Arch Intern Med* 2006;**166**:719–28.
16. Joshi S, Choi AD, Kamath GS, Raiszadeh F, Marrero D, Badheka A *et al.* Prevalence, predictors, and prognosis of atrial fibrillation early after pulmonary vein isolation: findings from 3 months of continuous automatic ECG loop recordings. *J Cardiovasc Electrophysiol* 2009;**20**:1089–94.
17. McCabe JM, Smith LM, Tseng ZH, Badhwar N, Lee BK, Lee RJ *et al.* Protracted CRP elevation after atrial fibrillation ablation. *Pacing Clin Electrophysiol* 2008;**31**:1146–51.
18. Yokokawa M, Tada H, Koyama K, Ino T, Naito S, Oshima S *et al.* Thickening of the left atrial wall shortly after radiofrequency ablation predicts early recurrence of atrial fibrillation. *Circ J* 2010;**74**:1538–46.
19. Hsieh MH, Chiou CW, Wen ZC, Wu CH, Tai CT, Tsai CF *et al.* Alterations of heart rate variability after radiofrequency catheter ablation of focal atrial fibrillation originating from pulmonary veins. *Circulation* 1999;**100**:2237–43.
20. Pappone C, Santinelli V, Manguso F, Vicedomini G, Gugliotta F, Augello G *et al.* Pulmonary vein denervation enhances long-term benefit after circumferential ablation for paroxysmal atrial fibrillation. *Circulation* 2004;**109**:327–34.
21. Lellouche N, Jais P, Nault I, Wright M, Bevilacqua M, Knecht S *et al.* Early recurrences after atrial fibrillation ablation: prognostic value and effect of early reablation. *J Cardiovasc Electrophysiol* 2008;**19**:599–605.
22. Deftereos S, Giannopoulos G, Kossyvakis C, Efremidis M, Panagopoulou V, Kaoukis A *et al.* Colchicine for prevention of early atrial fibrillation recurrence after pulmonary vein isolation: a randomized controlled study. *J Am Coll Cardiol* 2012;**60**:1790–6.
23. Aliot E, Capucci A, Crijns HJ, Goette A, Tamargo J. Twenty-five years in the making: flecainide is safe and effective for the management of atrial fibrillation. *Europace* 2011;**13**:161–73.
24. Kirchhof P, Andresen D, Bosch R, Borggrefe M, Meinertz T, Parade U *et al.* Short-term versus long-term antiarrhythmic drug treatment after cardioversion of atrial fibrillation (Flec-SL): a prospective, randomised, open-label, blinded endpoint assessment trial. *Lancet* 2012;**380**:238–46.
25. Kirchhof P, Engelen M, Franz MR, Ribbing M, Wasmer K, Breithardt G *et al.* Electrophysiological effects of flecainide and sotalol in the human atrium during persistent atrial fibrillation. *Basic Res Cardiol* 2005;**100**:112–21.
26. Calkins H, Reynolds MR, Spector P, Sondhi M, Xu Y, Martin A *et al.* Treatment of atrial fibrillation with antiarrhythmic drugs or radiofrequency ablation: two systematic literature reviews and meta-analyses. *Circ Arrhythm Electrophysiol* 2009;**2**:349–61.
27. Mun HS, Joung B, Shim J, Hwang HJ, Kim JY, Lee MH *et al.* Does additional linear ablation after circumferential pulmonary vein isolation improve clinical outcome in patients with paroxysmal atrial fibrillation? Prospective randomised study. *Heart* 2012;**98**:480–4.
28. Almroth H, Andersson T, Fengsrud E, Friberg L, Linde P, Rosenqvist M *et al.* The safety of flecainide treatment of atrial fibrillation: Long-term incidence of sudden cardiac death and proarrhythmic events. *J Intern Med* 2011;**270**:281–90.
29. Takada M, Goto T, Kotake T, Saito M, Kawato N, Nakai M *et al.* Appropriate dosing of antiarrhythmic drugs in japan requires therapeutic drug monitoring. *J Clin Pharm Ther* 2005;**30**:5–12.
30. Hori M, Connolly SJ, Zhu J, Liu LS, Lau CP, Pais P *et al.* Dabigatran versus warfarin: effects on ischemic and hemorrhagic strokes and bleeding in Asians and non-Asians with atrial fibrillation. *Stroke* 2013;**44**:1891–6.
31. Ogawa S, Shinohara Y, Kanmuri K. Safety and efficacy of the oral direct factor Xa inhibitor apixaban in Japanese patients with non-valvular atrial fibrillation—the ARISTOTLE-J study. *Circ J* 2011;**75**:1852–9.

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